

UNIT – 1 MINI PROJECT

SMART CAMPUS GROWTH & PLACEMENT PLATFORM

Detailed Unit–1 Product Engineering & Design Thinking Document

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Project Scope

This document presents the Smart Campus Growth & Placement Platform as a Unit–1 mini project. The first thirteen sections are intentionally expanded to provide detailed business understanding, design thinking, stakeholder analysis, requirements, product vision, MVP definition and roadmap information.

1. Project Overview

This section establishes the platform as a single student-development view, connecting academic progress, skills, achievements, mentoring and placement preparation so that development information can be understood as one continuous journey.

The Smart Campus Growth & Placement Platform (SCGPP) is proposed as a role-based campus web platform that connects student academic progress with skill development, achievements, mentoring and placement preparation. The main product idea is to avoid treating marks, attendance, projects, certificates and placement activities as isolated records. Instead, the platform creates one connected Student Growth Passport.

1.1 Background and Need

Students commonly participate in several development activities during their academic journey. Academic performance indicates one part of progress, while projects and certifications provide evidence of practical learning. Attendance can indicate participation, and mentoring can identify areas requiring attention. Placement preparation

adds another set of requirements. When these activities are separated, the student may have information but not a clear picture of overall readiness.

The proposed platform addresses this information gap by organizing development data around the student. It is intended to make information understandable and actionable rather than merely storing it.

1.2 Main Product Components

Component	Detailed Purpose
Student Growth Passport	Combines profile, academic summary, attendance, skills, projects, certificates and development status into one student-centered view.
Skill Matrix	Shows current skills, target skills, evidence and identified gaps so improvement can be planned.
Mentoring Module	Allows mentors to review assigned students, identify issues, record actions and follow up.
Readiness Module	Represents placement-preparation conditions as a visible checklist rather than an unexplained status.
Achievement Evidence	Connects projects, internships and certificates with relevant skills or development areas.
Reporting	Provides authorized summaries for students, mentors, placement staff and administrators.

1.3 Product Roles

Students use the platform to understand progress and complete development actions
Faculty mentors use assigned-student information to guide students
Placement officers use readiness and eligibility information for placement workflows
Administrators use controlled reports to understand department-level progress
IT/support personnel operate, secure and maintain the system.

1.4 Expected Product Outcome

The intended outcome is a campus workflow in which a student can answer four practical questions from the platform: Where am I now? What evidence do I have? What is missing? What should I do next? Mentors and placement staff should also receive role-appropriate information from the same underlying record.

2. Business Understanding

2.1 Business Problem

The business problem is fragmented student-development information. Academic records may exist in one location, attendance in another, project details in documents, certificates

in personal files and placement information in separate spreadsheets or processes. This fragmentation increases the effort required to understand the complete student profile.

The platform therefore begins with business understanding rather than implementation. The product must identify who uses the information, what each role needs, what decisions the information supports and which activities are repeated.

2.2 Stakeholder Needs

Stakeholder	Current Need / Problem	Business Need from Platform
Students	Do not always know which development activity should be done next.	Clear growth status, gaps, evidence and actions.
Faculty Mentors	Need to collect information before mentoring discussions.	Assigned-student summary and tracked action plans.
Placement Officers	Need reliable information for readiness and eligibility checks.	Consistent readiness view and supporting evidence.
Department Admin	Need department-level progress information.	Controlled reports and aggregated indicators.
Management	Needs an institutional view without unnecessary individual detail.	High-level trends and outcome reporting.
IT Support	Needs secure, maintainable operation.	Clear architecture, monitoring, backup and access control.

2.3 Business Objectives

Centralize relevant student-development information
Reduce repetitive manual compilation of progress and readiness information
Make skill gaps visible and connect them with actions
Make readiness requirements explainable
Support mentoring with structured student information
Provide controlled reporting for authorized roles
Create a scalable foundation for future campus-product features.

2.4 Business Rules to Validate

Which academic information should be included? Which skills are institutionally important? Who defines placement eligibility? Which evidence is accepted for a skill or achievement? Which roles may view or edit each information category? How long should records and evidence be retained?.

2.5 Business Success

Business success should be measured through reduced repeated work, improved visibility, clearer student actions and reliable role-specific information. Exact institutional measures should be validated with stakeholders before production deployment.

3. Design Thinking

The approach keeps the solution centered on actual student and stakeholder needs. Each stage helps reduce uncertainty before the next stage is developed, making the proposed product more practical and easier to validate.

Design Thinking is used to make the product user-centered. The project follows Empathize, Define, Ideate, Prototype and Test. Each stage reduces uncertainty before the next stage is expanded.

3.1 Empathize

Research Method	Participants	Questions / Learning
Student interviews	Students from different semesters	How do students track marks, projects, skills, certificates and placement preparation?
Mentor interviews	Faculty mentors	What information is needed before a mentoring session? What makes follow-up difficult?
Placement workflow study	Placement staff	How are eligibility and readiness currently checked? Where is information repeated?
Observation	Campus workflows	Which tasks involve repeated data entry, manual filtering or report preparation?
Document review	Forms, reports and records	What information is already collected and what information is missing?

3.2 Define

Core Problem Statement: Students have academic records, attendance, projects, certificates and placement activities scattered across different systems or files. Because there is no unified growth view, students may struggle to identify their gaps, mentors may spend time collecting information, and placement teams may repeat readiness and eligibility checks.

3.3 Ideate

Student Growth Passport Skill Matrix with current and target skills
Evidence-linked projects and certifications
Mentor action plans and follow-up dates
Placement-readiness checklist
Role-specific dashboards
Department-level reporting
Future QR/verified achievement sharing concept.

3.4 Prototype

The initial prototype should focus on the highest-value workflows: login, student dashboard, Growth Passport, Skill Matrix, mentor dashboard and readiness view. The prototype is not the final implementation; it is a tool for validating information structure and user understanding.

3.5 Test

Testing should examine whether users can find information, understand status labels, identify a skill gap, create or understand an action plan and interpret readiness requirements. Feedback should be recorded as observations rather than assumptions.

4. IBM Design Thinking

These principles help the team move from user needs to a clear product direction while keeping the proposed solution measurable, understandable and aligned with the campus problem described in the project.

4.1 Focus on User Outcomes

The product should be designed around outcomes rather than feature counts. A dashboard is valuable only when it helps a user understand or complete a task. A skill matrix is valuable when it helps identify a meaningful gap and supports a next action.

Product Element	User Outcome	Validation Question
Growth Passport	Student understands overall development.	Can the student explain current status after viewing it?
Skill Matrix	Skill gaps are visible.	Can the student identify which skill needs attention?
Evidence	Achievements are supported by records.	Can the user find evidence connected to a skill?
Mentor Action Plan	Guidance becomes trackable.	Can mentor and student see what action is pending?
Readiness View	Preparation requirements are transparent.	Can the user understand why a condition is incomplete?
Reports	Authorized staff obtain useful summaries.	Can the report answer the intended operational question?

4.2 Restless Reinvention

The product should continuously improve based on real usage. Feedback from students, mentors and placement staff should be converted into backlog items. If a feature creates complexity without measurable value, the team should reconsider its design.

4.3 Diverse Empowered Teams

Role	Responsibility
Business/Product Analyst	Requirements, domain understanding and stakeholder alignment
UI/UX Designer	Personas, journeys, prototypes and usability
Frontend Developer	Responsive user experience
Backend Developer	APIs, business logic and services
Database Engineer	Data model, integrity and performance
QA Engineer	Test strategy, validation and defect reporting
DevOps/IT	CI/CD, environments, deployment and monitoring

4.4 Team Working Principle

The team should be organized around shared product outcomes. Technical ownership remains clear, but major decisions should be evaluated by asking how they improve the user workflow.

5. IBM Loop Documentation

The loop encourages the team to observe actual needs, reflect on the evidence collected and make focused changes. This supports continuous learning instead of treating the first design as final.

The IBM Loop is represented as Observe → Reflect → Make. It provides a practical way to continue learning after the initial design has been created.

5.1 Observe in the SCGPP

Observe which dashboard areas users open most often
Observe where users stop or request help
Observe mentor time spent preparing for student reviews
Observe repeated placement eligibility checks
Observe incomplete skill evidence and unfinished action plans.

5.2 Reflect

Reflection converts observation into a product question. For example, if users do not update skills, the team should determine whether the screen is difficult, whether the skill

list is unclear, whether evidence is missing or whether the workflow has no perceived value.

5.3 Make

Make the smallest useful change, release it to the appropriate test group and observe the result. Examples include changing labels, simplifying a form, adding validation, improving search or restructuring a readiness checklist.

5.4 Loop Examples

Observation	Reflection	Product Change
Mentors repeatedly request the same student information.	Relevant information is distributed across screens.	Create a consolidated assigned-student summary.
Students ask why they are not ready.	Readiness status lacks explanation.	Show each incomplete requirement and supporting condition.
Skills are listed without evidence.	Skill level alone does not prove achievement.	Add evidence links and verification metadata.
Students have actions but forget follow-up.	Action plans lack visible due dates/status.	Add action status and follow-up dates.

5.5 Expected Benefit

The loop prevents the project from becoming a one-time document exercise. It provides a structured method for improving the product after prototype and MVP testing.

6. Stakeholder Analysis

Stakeholder analysis identifies people affected by the product, their influence, their information needs and the way they should participate in discovery and validation.

Stakeholder	Responsibilities	Needs	Influence	Engagement
Students	Use and maintain permitted development records	Simple, understandable growth view	High	Interviews, prototype tests, pilot
Faculty Mentors	Guide assigned students	Accurate summary and action tracking	High	Workflow workshops
Placement Officers	Manage readiness and placement activity	Eligibility, evidence and application information	High	Rule validation
Department Admin	Monitor department progress	Reports and controlled data	High	Dashboard review
Management	Review institutional outcomes	Aggregated indicators	High	Milestone demonstrations
IT Support	Operate and secure platform	Availability, backup, monitoring	Medium	Architecture/security review

6.1 Stakeholder Expectations

Students generally need simplicity and visibility. Mentors need relevant information without excessive administrative work. Placement staff need consistency and explainability. Administrators need useful reports. IT needs maintainability and secure operation.

6.2 Stakeholder Conflicts to Manage

A student may want maximum visibility while privacy rules require controlled access Management may want detailed reporting while individual information should not be exposed unnecessarily Placement staff may want strict eligibility rules while students need understandable explanations Mentors may want additional information that is not required for their assigned role.

6.3 Engagement Plan

Use interviews to discover needs Use prototypes to validate workflows Use acceptance criteria to validate functionality Use pilot feedback to identify improvement areas Document decisions when stakeholder requirements conflict.

7. User Personas

Personas represent the major users of the platform. They are used to design workflows around goals and frustrations rather than around database tables.

Persona	Goals	Frustrations	Platform Needs
Sakshi – Student	Become placement-ready and understand progress	Information is scattered; unclear next step	Growth Passport, Skill Matrix, actions, readiness
Mr. Patil – Mentor	Guide assigned students effectively	Manual collection before reviews	Assigned-student summary, gaps, action plans
Ms. Deshmukh – Placement Officer	Process eligible students and track activity	Manual filtering and verification	Readiness rules, evidence, application status
Dr. Joshi – Administrator	Monitor development trends	Reports require compilation	Aggregated analytics and controlled access

7.1 Student Persona – Sakshi

Sakshi has academic records, projects and certificates. She wants to know whether these activities are sufficient for her target placement opportunities. She needs a simple sequence from current status to gap to action. Her dashboard should avoid overwhelming her with administrative data.

7.2 Mentor Persona – Mr. Patil

Mr. Patil is responsible for a group of students. Before a meeting, he wants to see academic status, attendance concerns, skill gaps, recent achievements and previous action items. He needs to record decisions and follow-up dates so mentoring is continuous.

7.3 Placement Persona – Ms. Deshmukh

Ms. Deshmukh needs to review readiness and eligibility using information that can be traced to student records. She needs consistent rules and supporting evidence rather than informal judgments based on incomplete spreadsheets.

7.4 Administrator Persona – Dr. Joshi

Dr. Joshi needs a department or institutional view of development patterns. The dashboard should focus on aggregated information and avoid unnecessary exposure of personal details.

7.5 Persona Design Principle

Each persona should have a focused dashboard and task flow. The same underlying student-development data can support multiple roles, but the interface and permissions should reflect each role's responsibilities.

8. Stakeholder Responsibility Matrix

The responsibility matrix makes ownership explicit before implementation. It distinguishes who performs an activity, who approves or owns its outcome, who should be consulted and who should only be informed.

Activity	Student	Mentor	Placement	Admin	IT
Profile	R	C	C	A	S
Skills/Evidence	R	C	C	A	S
Mentoring	C	R/A	I	I	S
Readiness Rules	I	C	R/A	C	S
Reports	I	C	C	R/A	S
Security	I	I	I	C	R/A
Deployment	I	I	I	C	R/A
Testing	C	C	C	A	R

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed, S = Support.

8.1 Team Responsibility

Team	Primary Responsibilities	Expected Outputs
Business/Product	Problem, requirements, success criteria	Vision, requirements, acceptance criteria
Design	User research and experience	Personas, journeys, prototype
Engineering	Implementation	Frontend, backend, APIs, database
QA	Quality assurance	Test cases, defect reports, validation
DevOps/IT	Operations	CI/CD, environments, deployment, monitoring

8.2 Responsibility Flow

Identify requirement Confirm business owner Design and prototype Technical review
Implement Test Stakeholder validation Release and monitor.

8.3 Governance

Any requirement affecting privacy, access permissions, readiness rules or institutional reporting should be reviewed by the relevant business stakeholder before implementation.

9. Functional Requirements Specification (FRS)

The requirements provide a clear functional baseline for development and testing. Each workflow can be checked against the expected user action, validation and resulting system status.

Functional requirements describe what the system must do. The following requirements define the core behavior of the proposed MVP.

ID	Functional Requirement	Priority	Detailed Acceptance Condition
FR-01	Role-based authentication	Must	A valid user can authenticate and receives only the dashboard and functions permitted for the assigned role.
FR-02	Student profile and academics	Must	Authorized users can view/update permitted profile and academic information with validation.
FR-03	Attendance	Must	Attendance status is linked to the correct student and period and is displayed in the student context.
FR-04	Skill Matrix	Must	Skills can have current/target values and can be connected to evidence.
FR-05	Achievements	Must	Projects, internships and certificates can be recorded with relevant metadata.
FR-06	Evidence association	Must	Evidence can be linked to the appropriate skill or achievement and protected by access rules.

FR-07	Mentor assignment	Must	Mentors can see assigned students and record actions, notes and follow-up information.
FR-08	Readiness rules	Must	The system displays which readiness conditions are satisfied or incomplete.
FR-09	Placement activity	Should	Authorized placement users can track opportunity/application status.
FR-10	Reports	Should	Authorized users can generate defined student or department summaries.
FR-11	Audit history	Should	Important changes retain actor and timestamp information.

9.1 Functional Workflow

Authenticate user Load role-specific dashboard Retrieve only authorized information
 Perform permitted action Validate fields and relationships Save transaction Update
 dependent status where required Record audit information Display result and next action.

9.2 Validation Requirements

Required fields cannot be blank Controlled statuses must use predefined values Invalid
 references must be rejected Duplicate records should be controlled where business rules
 require uniqueness Evidence metadata should identify owner, type and related record
 Readiness results should be explainable.

9.3 Error Handling

The system should provide understandable validation messages, avoid exposing sensitive
 technical details to users and keep the application in a consistent state after a failed
 operation.

10. Non-Functional Requirements Specification (NFRS)

Non-functional requirements describe the quality characteristics expected from the
 product. They are important because a functionally correct campus system can still be
 unsuitable if it is insecure, difficult to use or unreliable.

ID	Requirement	Detailed Expectation
NFR-01	Security	Use role-based authorization, secure credential handling, protected APIs, validation and least-privilege access.
NFR-02	Performance	Common dashboard, search and status operations should respond quickly under expected academic/placement usage.

NFR-03	Availability	The platform should be dependable during academic and placement activity periods.
NFR-04	Usability	Screens should use clear labels, consistent navigation, readable status and responsive layouts.
NFR-05	Reliability	Use validation, constraints, transactions, backups and controlled error recovery.
NFR-06	Scalability	Architecture should support additional users, departments and modules.
NFR-07	Maintainability	Use modular code, documentation, testing and version control.
NFR-08	Privacy	Users should only access information required by their role and workflow.

10.1 Security Controls

Authentication and authorization must be separated conceptually Permissions must be checked at backend/API level, not only by hiding interface buttons Credentials and secrets should not be stored in source code Evidence files should have controlled access Important changes should be auditable Input should be validated before processing.

10.2 Privacy and Data Governance

The platform handles student-development information, so data governance must be designed into the product. The team should define ownership, permitted access, retention needs, correction procedures and the minimum information necessary for each workflow.

10.3 Reliability and Recovery

Database constraints prevent invalid relationships Transactions keep related operations consistent Backups support recovery Monitoring identifies operational failures Controlled error messages support recovery without revealing sensitive implementation details.

10.4 Maintainability

The application should be modular enough that a change to readiness rules, for example, does not require unrelated student-profile components to be rewritten.

11. Product Vision

11.1 Vision Statement

To provide a unified digital Student Growth Passport that connects academic progress, attendance, skills, achievements, mentoring and placement preparation so every student can clearly understand and improve their readiness for future opportunities.

11.2 Mission Statement

Make student development measurable, visible and actionable by connecting fragmented academic and career-readiness information in one secure platform.

11.3 Product Objectives

Reduce manual progress and readiness reporting Give students a single place to understand development status Improve mentor visibility into assigned students Create consistent and configurable readiness checks Provide authorized management users with useful aggregated indicators Create a foundation for future campus-product expansion.

11.4 Product Promise

The product should help a student understand current position, evidence, missing requirements and next actions. It should help mentors convert identified gaps into tracked actions and help placement staff work with consistent readiness information.

11.5 Success Criteria

Measure	What Success Would Look Like
Profile completeness	Active student profiles contain the core information needed by defined workflows.
Mentoring efficiency	Mentors can review assigned students without manually collecting the same information.
Readiness transparency	Students can understand incomplete requirements.
Reporting effort	Routine summaries can be generated from the platform.
Adoption	Pilot users return to the core workflow and complete intended tasks.

11.6 Product Principles

Make progress visible Make gaps actionable Keep evidence connected Support mentors rather than replace them Protect role-specific information Iterate from real usage.

12. Minimum Viable Product (MVP)

The MVP is intentionally limited to the workflows needed to validate the central product idea. It should be small enough to build and test while still representing the Student Growth Passport concept.

Module	MVP Scope	Purpose	Primary User
Authentication	Student, Mentor, Placement Officer and Admin roles	Secure role-based access	All
Growth Passport	Profile, academics, attendance, skills and achievements	Create connected student view	Student
Skill Matrix	Current level, target level, evidence	Make gaps visible	Student/Mentor
Mentoring	Assigned students, notes, actions, follow-up	Turn gaps into actions	Mentor
Readiness	Rule-based checklist and explanations	Make preparation transparent	Student/Placement
Basic Reporting	Student progress and department summaries	Operational visibility	Admin/Placement

12.1 MVP Acceptance Criteria

All supported roles can authenticate
A student can view the Growth Passport
A student can maintain permitted skills and achievements
A mentor can review an assigned student
A mentor can create/update an action plan
Readiness status shows incomplete conditions
Authorized users can generate basic reports.

12.2 Out of Scope for First Release

AI career recommendations
Predictive placement probability
External job-board integrations
Automated resume generation
Advanced third-party assessment integrations
Complex predictive analytics.

12.3 MVP Learning Questions

Do students understand the Growth Passport? Can users identify skill gaps? Can mentors conduct reviews with less preparation effort? Do users understand why readiness conditions are incomplete? Does the common data model support the intended workflows?.

12.4 MVP Principle

The MVP should validate the product concept before adding advanced automation. Expansion should be based on evidence from real or controlled pilot usage.

13. Product Roadmap

The roadmap provides a staged path from discovery and design to implementation, validation and future expansion, allowing the team to learn from each phase before increasing scope.

The roadmap sequences the product from discovery to implementation, quality and future expansion. It should be treated as a learning-oriented plan rather than a rigid list of features.

Phase	Focus	Detailed Outputs	Exit Evidence
Phase 1	Domain Analysis	Problem, stakeholders, personas, requirements	Stakeholder validation
Phase 2	Design Thinking	Journeys, wireframes, prototype, test feedback	Usability findings
Phase 3	MVP Development	Authentication, passport, skills, mentoring, readiness, reporting	Working MVP + tests
Phase 4	Quality & DevOps	Automation, CI/CD, deployment, monitoring, security checks	Release-ready build
Phase 5	Expansion	Placement applications, advanced reporting, integrations	Validated demand
Phase 6	Continuous Improvement	Usage analytics, feedback, iterative releases	Measured improvements

13.1 Agile Working Model

Maintain a prioritized product backlog
Convert requirements into small user stories
Define acceptance criteria before implementation
Design and develop in increments
Review completed work with stakeholders
Track defects separately from future enhancements
Use retrospectives to improve the development process.

13.2 Release Gate

A release should move forward when relevant acceptance criteria, basic security checks, data validation and critical workflow tests are complete. A feature can be postponed if it increases complexity without helping validate the core product outcome.

13.3 Roadmap Risks

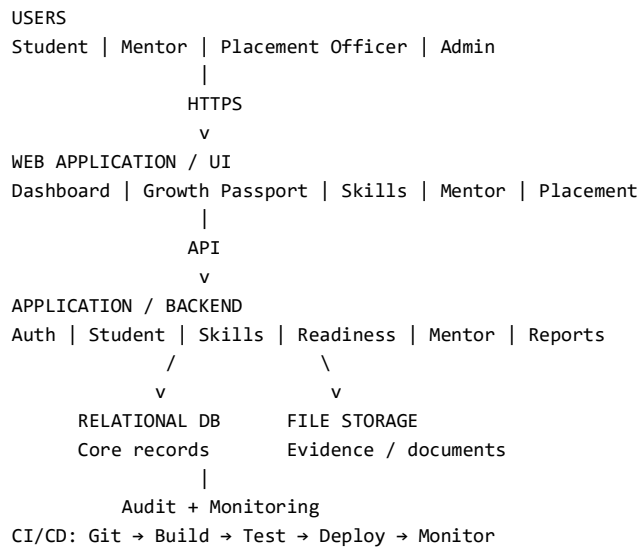
Requirements may change after user research
Institutional readiness rules may differ
Data quality may be inconsistent
Additional integrations may increase scope
Users may need training or onboarding.

13.4 Product Learning Cycle

BUILD → MEASURE → LEARN. Every phase should generate evidence that informs the next phase. This approach keeps the roadmap connected to user outcomes.

14. Initial System Architecture Diagram

The architecture separates major responsibilities so that the interface, business logic, data and supporting services can evolve without making the whole platform difficult to maintain.



The architecture separates interface, application logic and storage. Authorization should be enforced at the service/API layer. Evidence storage should have controlled access. The design should remain modular so future placement, analytics or integration modules can be added without redesigning the entire system.

Entity	Purpose
User	Identity and role
Student	Profile and department
Academic Record	Semester information
Attendance	Attendance status
Skill	Current/target capability
Evidence	Supporting achievement record
Mentor Action	Action and follow-up
Readiness Rule	Requirement condition
Placement Activity	Opportunity/application status
Audit Log	Change history

15. Project Repository Initialization

Using a shared repository also improves traceability because requirements, implementation changes and testing work can be reviewed as part of the same project history.

The repository is the shared technical and documentation workspace. It should make the project traceable from requirements to implementation and testing.

```
git init
git add .
git commit -m "Initialize Unit 1 product discovery and project structure"
git branch -M main
```

15.1 Branch Structure

main – stable project version develop – integration branch feature/* – individual feature work fix/* – defect correction.

15.2 Repository Practices

Use clear commit messages Commit coherent changes Do not commit credentials or secrets Review changes before merging Keep major documentation changes synchronized with implementation.

15.3 Unit-1 Contents

Business understanding Stakeholders and personas FRS/NFRS Design Thinking MVP Roadmap Architecture Charter README and documentation.

16. Project Folder Structure

The folder structure keeps related application, documentation, deployment and testing resources organized, making the project easier for team members to understand and maintain.

```
smart-campus-growth-placement-platform/
├── .github/workflows/
├── app/
│   ├── frontend/
│   ├── backend/
│   ├── modules/
│   │   ├── auth/
│   │   ├── students/
│   │   ├── academics/
│   │   ├── skills/
│   │   ├── mentoring/
│   └── placement/
```

```

|   | database/
|   | tests/
|   |
|   | docs/
|   | deployment/
|   | monitoring/
|   | README.md
|   | LICENSE

```

Folder	Purpose
frontend	User interface
backend	APIs and business logic
modules	Feature-specific services
database	Schema/migrations
tests	Automated testing
docs	Project documentation
deployment	Environment/deployment
monitoring	Logs and monitoring

The structure keeps product features, documentation, deployment and testing separated so that later development can scale without becoming difficult to maintain.

17. Project Charter

The charter provides a compact reference for the project's purpose, users, scope, risks and expected success, helping the team keep later implementation aligned with the original objective.

Item	Definition
Project	Smart Campus Growth & Placement Platform
Prepared By	Saksham Kurhe
Core Concept	Student Growth Passport
Problem	Fragmented development information
Goal	Make growth measurable, visible and actionable
Primary Users	Students, Mentors, Placement Officers, Administrators
MVP	Authentication, passport, skills, mentoring, readiness, reporting
Risks	Data quality, privacy, inconsistent rules, adoption
Success	Users understand and act on development status

17.1 Scope

The project focuses on the first usable foundation and does not attempt to automate every campus process in the initial release.

17.2 Constraints

Institutional policies may differ Student information requires controlled access Readiness rules need stakeholder validation MVP scope must remain controlled Adoption depends on usability and data completeness.

18. Product Discovery Documentation

The discovery documentation records the main decisions and artifacts produced during Unit 1 and provides a reference for validating the product before development scope is expanded.

Artifact	Purpose	Status
Business Understanding	Problem and domain	Prepared
Stakeholder Analysis	Users and decision makers	Prepared
Personas	User needs	Prepared
Design Thinking	Research and solution process	Prepared
FRS/NFRS	Functional and quality requirements	Prepared
Product Vision	Direction and principles	Prepared
MVP	First-release scope	Prepared
Roadmap	Development sequence	Prepared
Architecture	Technical foundation	Prepared
Project Charter	Scope and risks	Prepared

18.1 Discovery Principle

The project starts with user and business understanding rather than coding. Discovery artifacts should be reviewed with representative users before the product scope is expanded.

18.2 Evidence to Collect Next

Interview notes Prototype usability observations Readiness-rule validation MVP acceptance feedback Pilot usage metrics.

19. Initial User Journey

The journey shows how the different modules work together from login and review through action, evidence, mentoring and readiness updates, giving the team a common workflow to validate.

Stage	Student	Mentor	Placement Officer
Access	Login and dashboard	Login and assigned students	Login and readiness
Review	Academics, attendance, skills	Student summary	Readiness information

Identify	Gaps and missing evidence	Development gaps	Incomplete requirements
Act	Complete tasks	Create/follow action plan	Manage eligibility/application
Validate	Review updated status	Review completion	Review authorized evidence
Progress	Readiness updates	Follow up	Use updated information

19.1 End-to-End Sequence

Student logs in Student reviews Growth Passport Student updates skill/evidence information Platform highlights gaps Mentor reviews and records actions Student completes development tasks Readiness checklist updates Placement officer uses authorized readiness information.

The journey is intentionally continuous: information is not collected only for a report; it is used to guide the next student action.

20. Key Product Differentiator & Unit-1 Deliverables

The differentiating idea is the connected Student Growth Passport: academic information, attendance, skills, projects, certifications, mentoring and placement preparation are brought together so that gaps can lead to clear next actions.

20.1 Student Growth Passport

The key differentiator is the connection between student information and student growth. Academic records, attendance, skills, projects, certificates, mentoring actions and readiness conditions can contribute to one evolving development view.

20.2 Unit-1 Deliverables

☒ Business Understanding

☒ Product Vision

☒ Stakeholder Analysis

☒ User Personas

☒ FRS

☒ NFRS

- ☑ Design Thinking
- ☑ IBM Design Thinking
- ☑ IBM Loop Documentation
- ☑ MVP Definition
- ☑ Product Roadmap
- ☑ Initial Architecture
- ☑ Project Charter
- ☑ Repository Initialization
- ☑ Folder Structure
- ☑ Product Discovery Documentation.

20.3 Conclusion

The Smart Campus Growth & Placement Platform establishes a structured Unit-1 foundation for a campus product. The first thirteen sections provide detailed product reasoning, requirements and planning, while the remaining sections connect those decisions to architecture, repository structure, charter, discovery, user journey and deliverables.